

India Air Pollution Analysis using Python

The objective of this project is to analyze air pollution monitoring data across India, identify pollution patterns, compare states and cities, and derive meaningful insights through exploratory data analysis.

1: Data Understanding

1: Import Libraries

```
In [42]: import numpy as np  
import pandas as pd  
import matplotlib.pyplot as plt  
import seaborn as sns
```

2: Load Dataset

```
In [43]: aqi = pd.read_csv("India Air Pollution.csv")
```

3: Dataset Overview

In [44]: `aqi.head()`

Out[44]:

	country	state	city	station	last_update	latitude
0	India	Bihar	Patna	IGSC Planetarium Complex, Patna - BSPCB	02-07-2026 14:00:00	25.610369
1	India	Bihar	Patna	IGSC Planetarium Complex, Patna - BSPCB	02-07-2026 14:00:00	25.610369
2	India	Bihar	Purnia	Mariam Nagar, Purnia - BSPCB	02-07-2026 14:00:00	25.366336
3	India	Bihar	Rajgir	Dangi Tola, Rajgir - BSPCB	02-07-2026 14:00:00	25.032800
4	India	Bihar	Saharsa	Police Line, Saharsa - BSPCB	02-07-2026 14:00:00	25.892357

In [45]: `aqi.info()`

```
<class 'pandas.DataFrame'>
RangeIndex: 3514 entries, 0 to 3513
Data columns (total 11 columns):
#   Column          Non-Null Count  Dtype
---  -
0   country         3514 non-null   str
1   state           3514 non-null   str
2   city            3514 non-null   str
3   station         3514 non-null   str
4   last_update     3514 non-null   str
5   latitude        3514 non-null   float64
6   longitude       3514 non-null   float64
7   pollutant_id    3514 non-null   str
8   pollutant_min   3158 non-null   float64
9   pollutant_max   3158 non-null   float64
10  pollutant_avg    3158 non-null   float64
dtypes: float64(5), str(6)
memory usage: 559.9 KB
```

In [46]: `aqi.describe()`

Out[46]:

	latitude	longitude	pollutant_min	pollutant_max	po
count	3514.000000	3514.000000	3158.000000	3158.000000	3158.000000
mean	23.277241	78.534398	16.266941	46.640279	28.266941
std	5.155843	4.874285	16.268631	59.296273	27.268631
min	8.514909	70.776774	0.000000	0.000000	0.000000
25%	19.215784	75.387310	5.000000	13.000000	9.000000
50%	23.811588	77.272783	11.000000	29.000000	20.000000
75%	28.060249	80.612222	23.000000	57.000000	37.000000
max	34.066206	94.636574	134.000000	500.000000	295.000000

In [47]: `aqi.shape`

Out[47]: (3514, 11)

Observation:

- The dataset contains air pollution records collected from monitoring stations across India.
- It includes location information such as state, city, station, latitude, and longitude.
- Pollution measurements include minimum, maximum, and average values for different pollutants.
- The dataset also contains the timestamp of the latest observation.

4: Data Cleaning

In [48]: `aqi.isnull().sum()`

Out[48]:

```
country      0
state        0
city         0
station      0
last_update  0
latitude     0
longitude    0
pollutant_id 0
pollutant_min 356
pollutant_max 356
pollutant_avg 356
dtype: int64
```

```
In [49]: (aqi.isnull().sum()/len(aqi))*100
```

```
Out[49]: country          0.000000  
         state            0.000000  
         city             0.000000  
         station          0.000000  
         last_update      0.000000  
         latitude         0.000000  
         longitude        0.000000  
         pollutant_id     0.000000  
         pollutant_min    10.130905  
         pollutant_max    10.130905  
         pollutant_avg    10.130905  
         dtype: float64
```

```
In [50]: aqi = aqi.dropna(subset=[  
         "pollutant_min",  
         "pollutant_max",  
         "pollutant_avg"  
         ])
```

```
In [51]: aqi.isnull().sum()
```

```
Out[51]: country          0  
         state            0  
         city             0  
         station          0  
         last_update      0  
         latitude         0  
         longitude        0  
         pollutant_id     0  
         pollutant_min    0  
         pollutant_max    0  
         pollutant_avg    0  
         dtype: int64
```

```
In [52]: aqi.duplicated().sum()
```

```
Out[52]: np.int64(0)
```

```
In [53]: # checking data type
         aqi.dtypes
```

```
Out[53]: country          str
         state            str
         city             str
         station          str
         last_update      str
         latitude         float64
         longitude        float64
         pollutant_id     str
         pollutant_min    float64
         pollutant_max    float64
         pollutant_avg    float64
         dtype: object
```

5. Convert Date Column

```
In [54]: aqi["last_update"] = pd.to_datetime(aqi["last_update"])
```

```
In [55]: aqi.dtypes
```

```
Out[55]: country          str
         state            str
         city             str
         station          str
         last_update      datetime64[us]
         latitude         float64
         longitude        float64
         pollutant_id     str
         pollutant_min    float64
         pollutant_max    float64
         pollutant_avg    float64
         dtype: object
```

6. Create New Time Features

```
In [56]: aqi["year"] = aqi["last_update"].dt.year
```

```
In [57]: aqi["month"] = aqi["last_update"].dt.month
```

```
In [58]: aqi["day"] = aqi["last_update"].dt.day
```

```
In [59]: aqi["hour"] = aqi["last_update"].dt.hour
```

```
In [60]: aqi["day_name"] = aqi["last_update"].dt.day_name()
```

Observation:

- The dataset contained missing values in pollution measurement columns.
- Missing rows were removed to ensure accurate analysis.
- The date column was converted to datetime format for time-based analysis.

2: Exploratory Data Analysis (EDA)

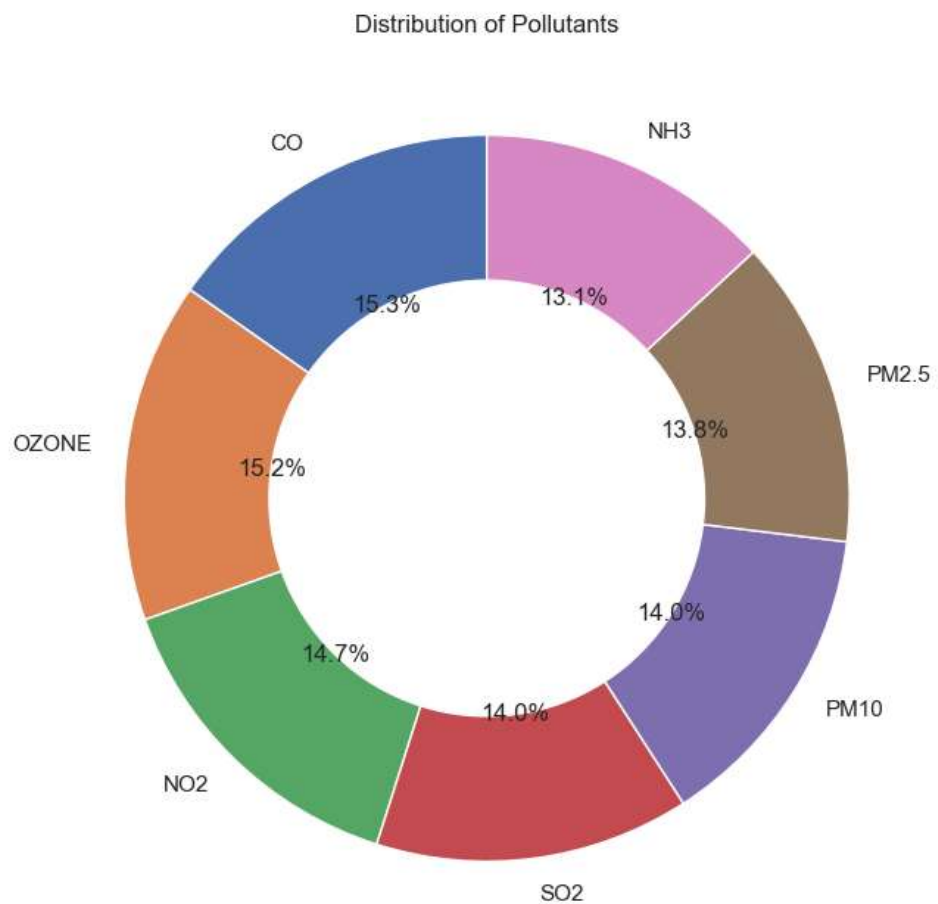
1. Distribution of Air Pollutants Across Monitoring Stations

```
In [61]: pollutant_counts = aqi["pollutant_id"].value_counts()

plt.figure(figsize=(8,8))

plt.pie(
    pollutant_counts,
    labels=pollutant_counts.index,
    autopct="%1.1f%%",
    startangle=90,
    wedgeprops=dict(width=0.4)
)

plt.title("Distribution of Pollutants")
plt.show()
```



Observation

Carbon Monoxide (CO) has the highest number of recorded observations in the dataset.

This indicates that CO is monitored more frequently than the other pollutants.

Pollutants with fewer records may have limited monitoring coverage across monitoring stations.

2. Which states contribute the most pollution monitoring records?

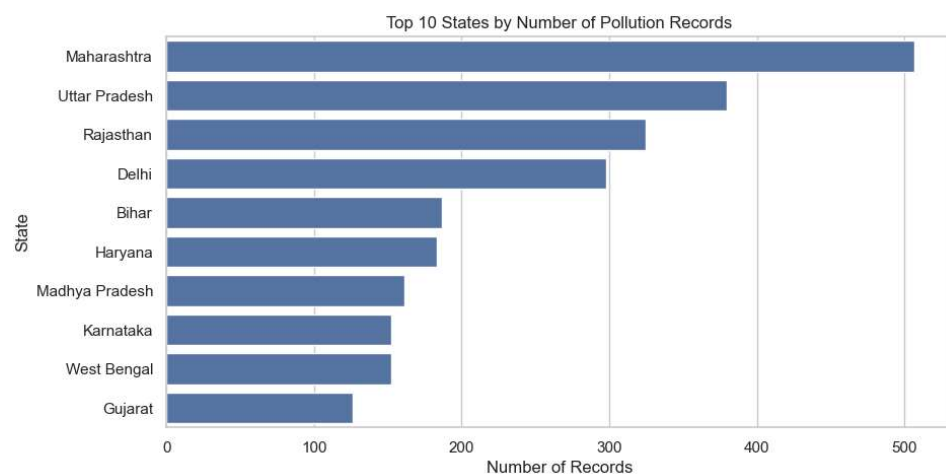
```
In [62]: top_states = aqi["state"].value_counts().head(10)

plt.figure(figsize=(10,5))

sns.barplot(
    x=top_states.values,
    y=top_states.index
)

plt.title("Top 10 States by Number of Pollution Records")
plt.xlabel("Number of Records")
plt.ylabel("State")

plt.show()
```



Observation

Maharashtra has the highest number of pollution monitoring records.

This suggests that Maharashtra has a wider monitoring network or more frequent data collection compared to other states.

The number of records reflects monitoring activity and should not be interpreted as the pollution level.

3. Which cities have the most pollution monitoring records?


```

In [63]: top_cities = aqi["city"].value_counts().head(10)

plt.figure(figsize=(10,6))

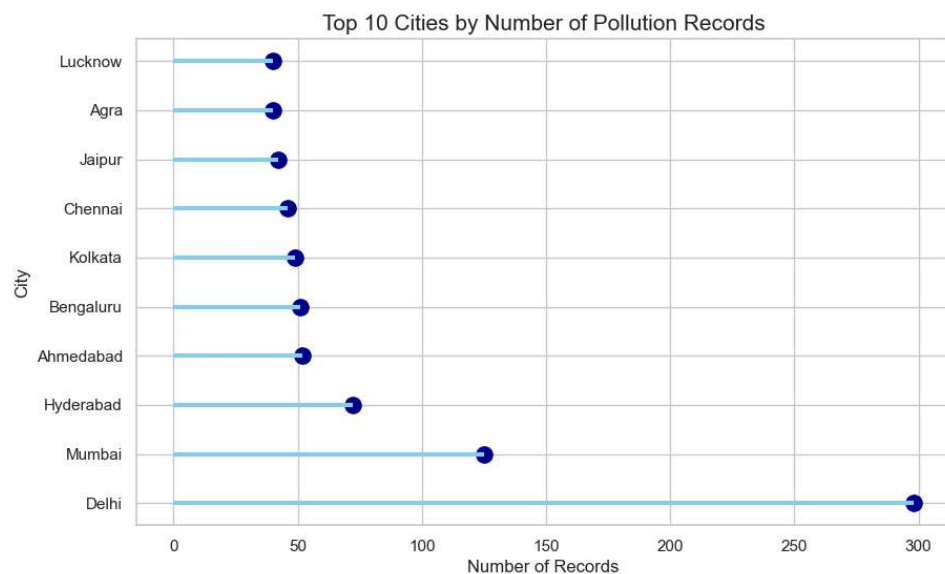
# Draw lines
plt.hlines(
    y=top_cities.index,
    xmin=0,
    xmax=top_cities.values,
    color="skyblue",
    linewidth=3
)

# Draw points
plt.scatter(
    top_cities.values,
    top_cities.index,
    color="darkblue",
    s=120
)

plt.title("Top 10 Cities by Number of Pollution Records", fontsize=15)
plt.xlabel("Number of Records")
plt.ylabel("City")

plt.show()

```



Obervation

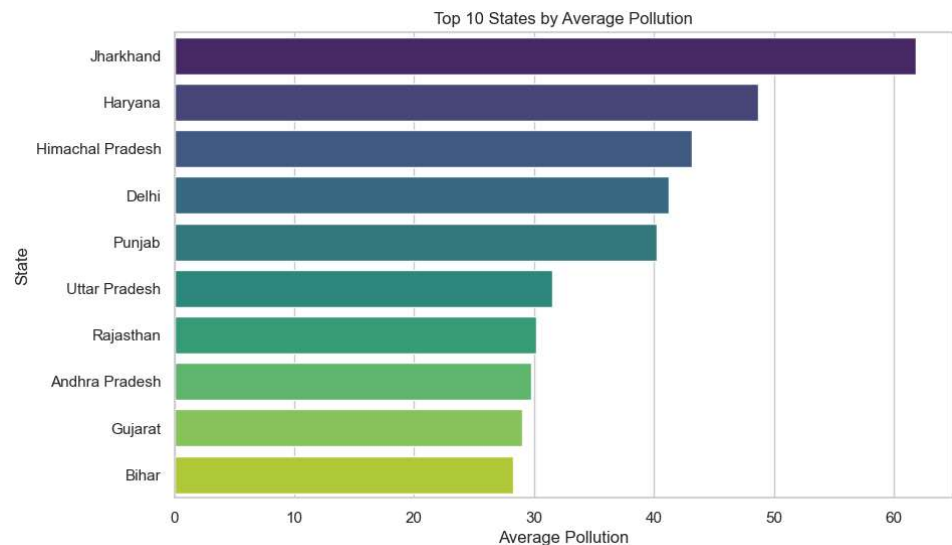
Delhi has the highest number of pollution records, followed by Mumbai.

These cities appear to have extensive air quality monitoring infrastructure.

A higher number of observations improves the reliability of pollution analysis for these locations.

4. Top 10 Indian States by Average Pollution Level

```
In [64]: state_pollution = (  
    aqi.groupby("state")["pollutant_avg"]  
        .mean()  
        .sort_values(ascending=False)  
        .head(10)  
    )  
  
    plt.figure(figsize=(10,6))  
  
    sns.barplot(  
        x=state_pollution.values,  
        y=state_pollution.index,  
        palette="viridis"  
    )  
  
    plt.title("Top 10 States by Average Pollution")  
  
    plt.xlabel("Average Pollution")  
  
    plt.ylabel("State")  
  
    plt.show()
```



Observation:

Jharkhand records the highest average pollution level among all states in the dataset.

This suggests that monitoring stations in Jharkhand experienced relatively higher pollutant concentrations.

Unlike the previous chart, this visualization represents pollution intensity rather than the number of monitoring records.

5: Top 10 Most Polluted Cities

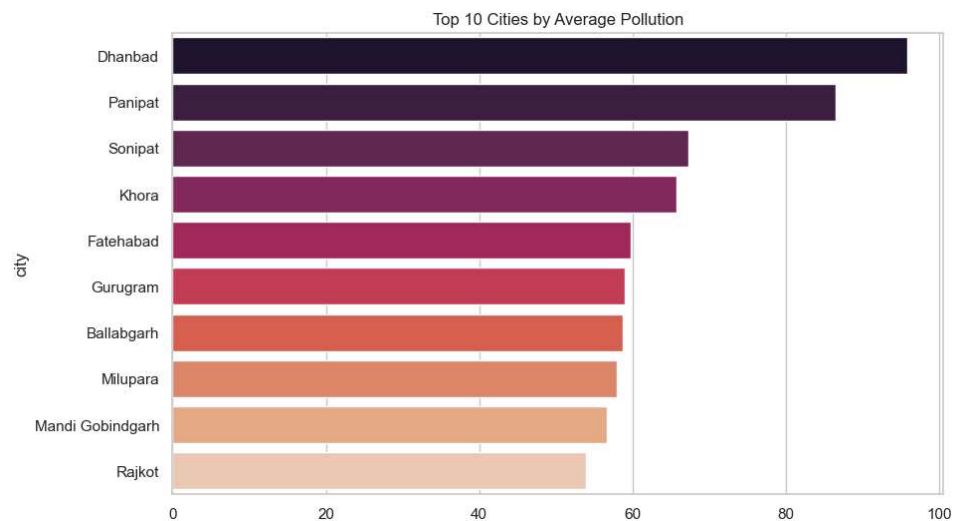
```
In [65]: city_pollution = (
    aqi.groupby("city")["pollutant_avg"]
        .mean()
        .sort_values(ascending=False)
        .head(10)
    )

plt.figure(figsize=(10,6))

sns.barplot(
    x=city_pollution.values,
    y=city_pollution.index,
    palette="rocket"
)

plt.title("Top 10 Cities by Average Pollution")

plt.show()
```



Observation:

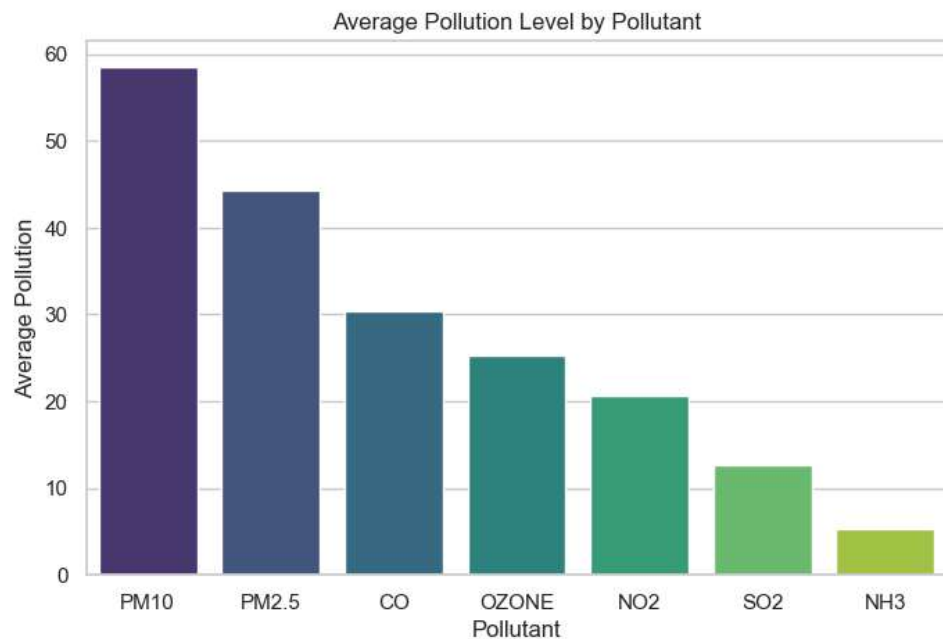
Dhanbad has the highest average pollution level, followed by Panipat.

These cities recorded higher pollutant concentrations than other cities in the dataset.

The findings indicate that these locations may require stronger pollution control and environmental monitoring measures.

6: Average Pollution by Pollutant Type

```
In [66]: pollutant_avg = (  
    aqi.groupby("pollutant_id")["pollutant_avg"]  
        .mean()  
        .sort_values(ascending=False)  
    )  
  
    plt.figure(figsize=(8,5))  
  
    sns.barplot(  
        x=pollutant_avg.index,  
        y=pollutant_avg.values,  
        palette="viridis"  
    )  
  
    plt.title("Average Pollution Level by Pollutant")  
  
    plt.xlabel("Pollutant")  
  
    plt.ylabel("Average Pollution")  
  
    plt.show()
```



Observation:

PM10 has the highest average concentration among all monitored pollutants.

NH3 (Ammonia) has the lowest average concentration in the dataset.

The higher concentration of particulate matter highlights it as one of the major contributors to air pollution.

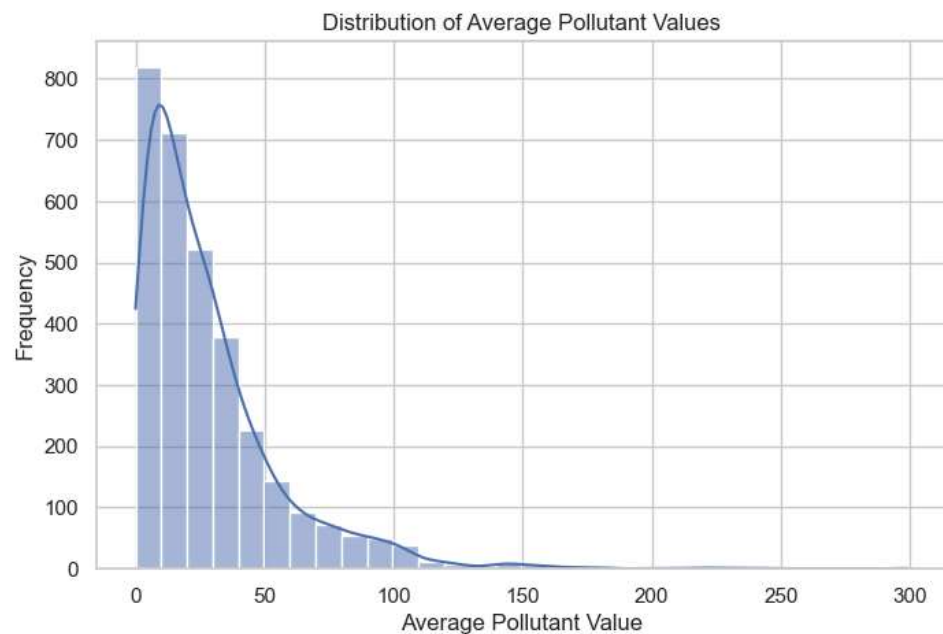
Distribution of Average Pollutant Values

```
In [67]: plt.figure(figsize=(8,5))

sns.histplot(aqi["pollutant_avg"], bins=30, kde=True)

plt.title("Distribution of Average Pollutant Values")
plt.xlabel("Average Pollutant Value")
plt.ylabel("Frequency")

plt.show()
```



Observation

Most pollution measurements are concentrated in the lower to moderate range.

A small number of observations have very high pollutant values, creating a right-skewed distribution.

These high-value observations may represent severe pollution episodes that warrant further investigation.

Key Findings

The dataset contains air pollution records collected from monitoring stations across India.

Carbon Monoxide (CO) is the most frequently monitored pollutant in the dataset.

Maharashtra has the highest number of pollution monitoring records, indicating extensive monitoring coverage.

Delhi records the highest number of city-level observations, followed by Mumbai.

Jharkhand has the highest average pollution level among all states.

Dhanbad and Panipat are the two cities with the highest average pollutant concentrations.

PM10 has the highest average pollutant concentration, while NH3 has the lowest.

Most pollutant measurements fall within lower concentration ranges, with a few extreme values indicating severe pollution events.

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